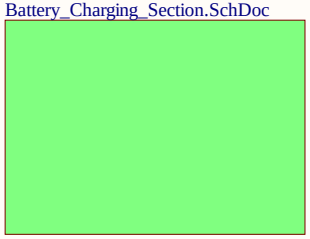
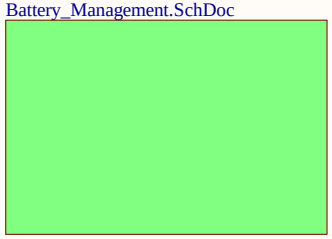
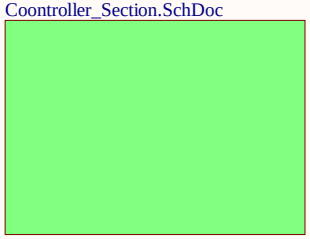
U_Battery_Charging
Battery_Charging_Section.SchDoc



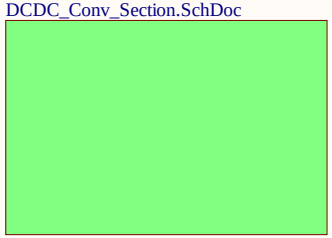
U_Battery_Management
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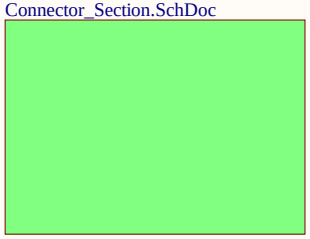
U_Conntroller_Section
Coontroller_Section.SchDoc



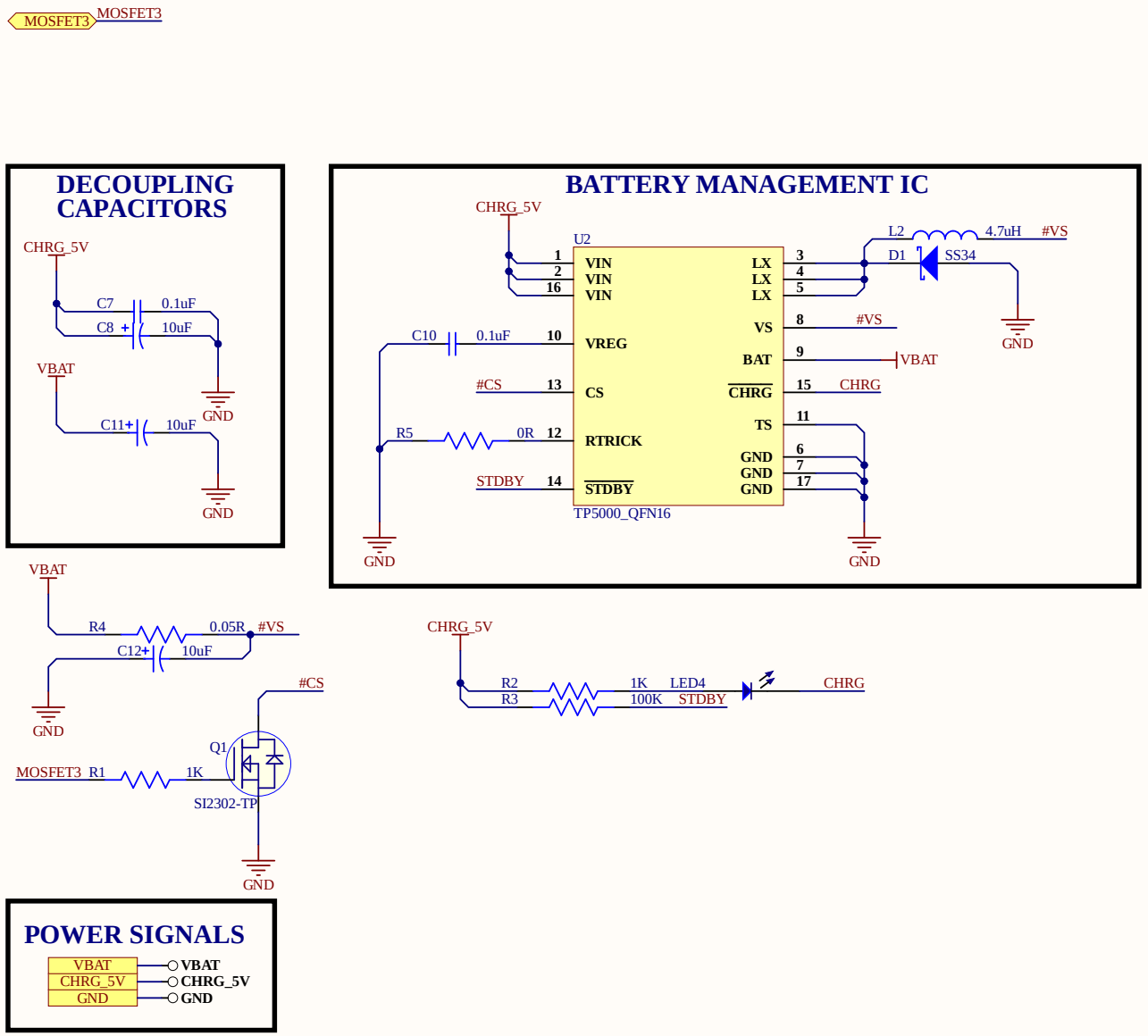
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DCDC_Conv_Section.SchDoc

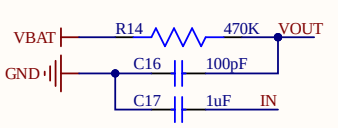


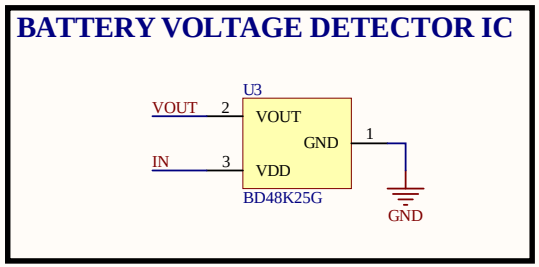
U_Connector_Section
Connector_Section.SchDoc

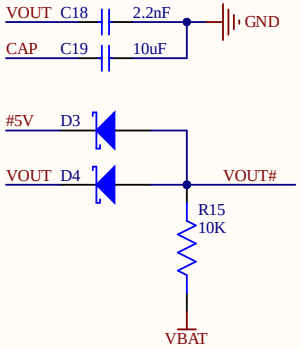


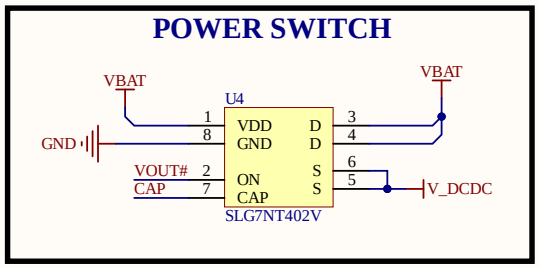
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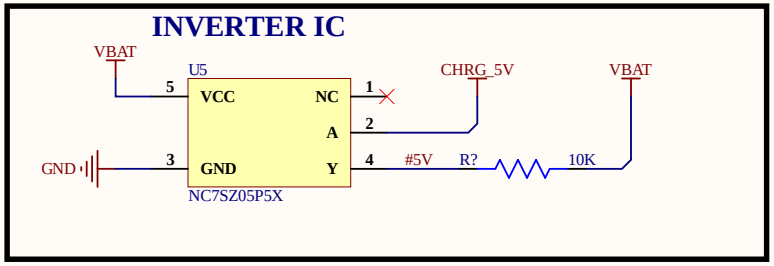

MOSFET3
MOSFET3
DECOUPLING CAPACITORS
CHRG_5V
C7
0.1uF
C8
10uF
VBAT
GND
C11
10uF
GND
BATTERY MANAGEMENT IC
CHRG_5V
U2
1 VIN
2 VIN
16 VIN
10 VREG
13 CS
12 RTRICK
14 STDBY
3 LX
4 LX
5 LX
8 VS
9 BAT
15 CHRG
11 TS
6 GND
7 GND
17 GND
TP5000_QFN16
L2
4.7uH
#VS
D1
SS34
GND
#VS
VBAT
CHRG
C10
0.1uF
R5
0R
#CS
STDBY
GND
GND
VBAT
R4
0.05R
#VS
C12
10uF
GND
#CS
Q1
MOSFET3_R1
1K
SI2302-TP
GND
CHRG_5V
R2
1K
LED4
CHRG
R3
100K
STDBY
POWER SIGNALS
VBAT
CHRG_5V
GND
VBAT
CHRG_5V
GND

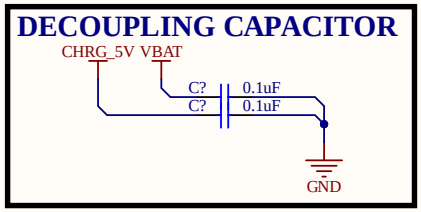


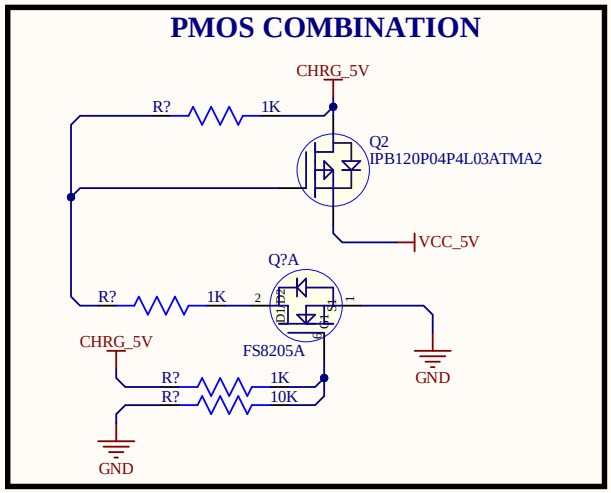


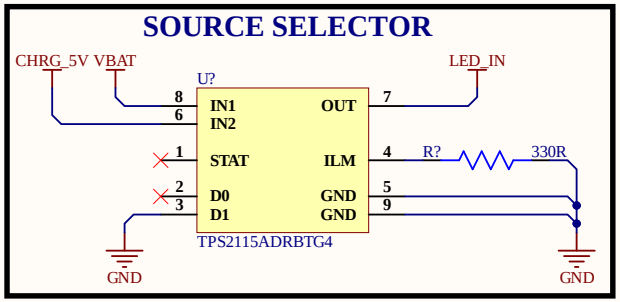












STAT pin can be pulled up to output and connected to MCU for getting IC status

IL=500/RILM

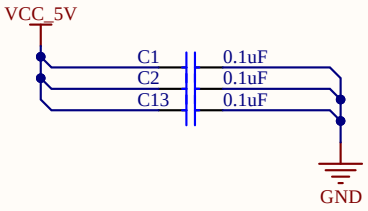
A1 — A1

MOSFET3 — MOSFET3

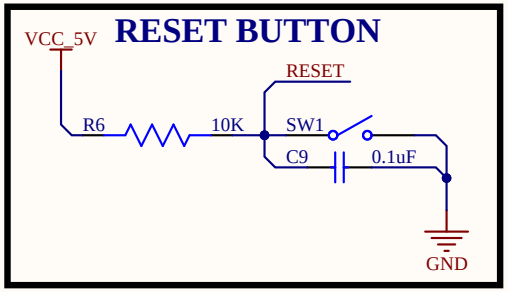
ISP SIGNALS



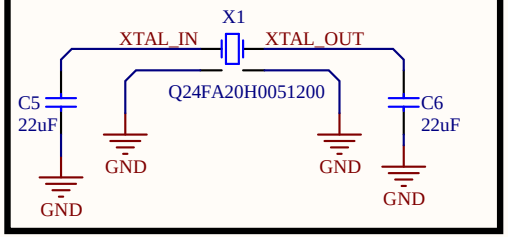
DECOUPLING CAPACITORS



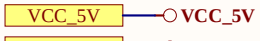
RESET BUTTON



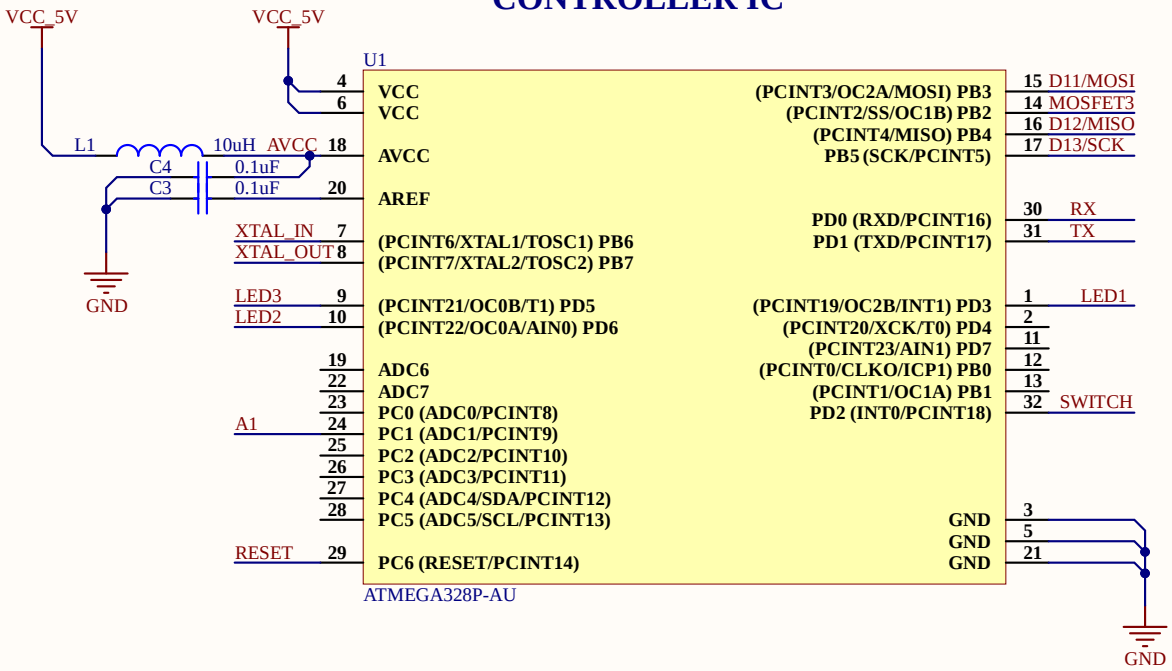
16MHz CRYSTAL



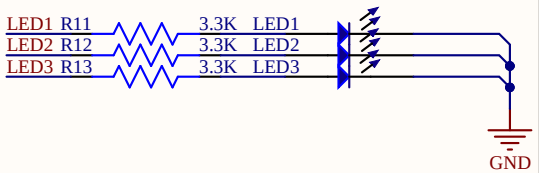
POWER SIGNALS



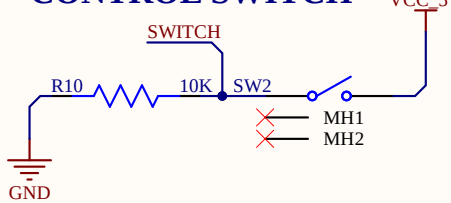
CONTROLLER IC



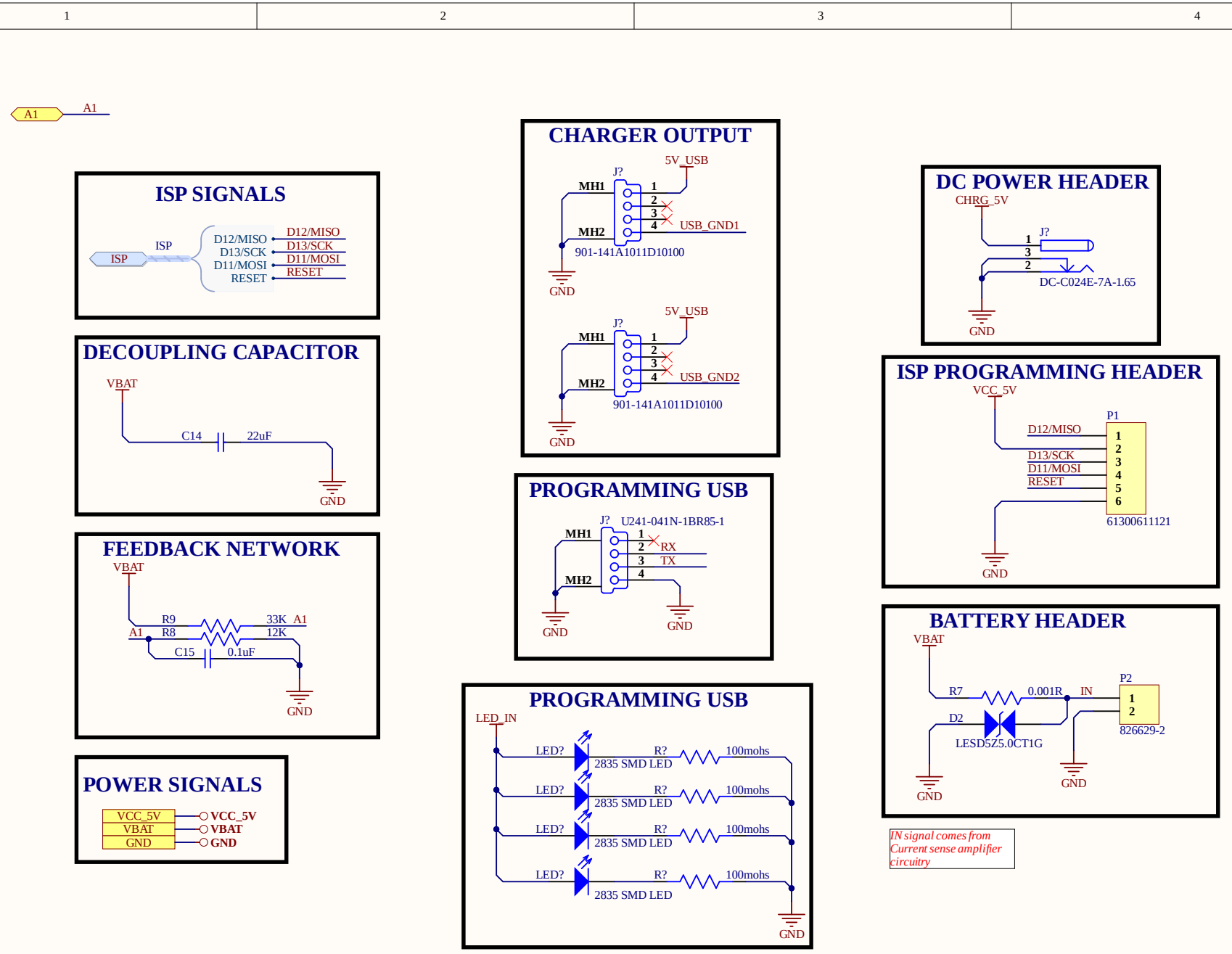
INDICATION LEDS



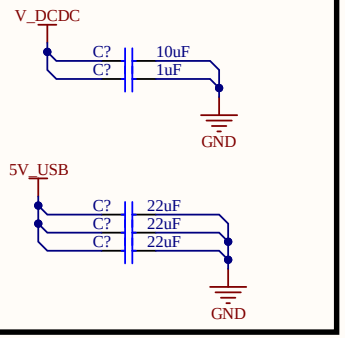
CONTROL SWITCH



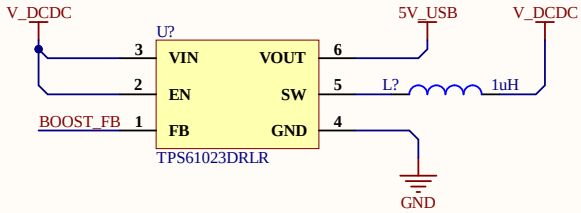
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File: D:\Github\..\Coontroller_Section.SchDoc		Drawn By:


A1
A1
ISP SIGNALS
ISP
D12/MISO
D13/SCK
D11/MOSI
RESET
D12/MISO
D13/SCK
D11/MOSI
RESET
DECOUPLING CAPACITOR
VBAT
C14
22uF
GND
FEEDBACK NETWORK
VBAT
R9
33K
A1
A1
R8
12K
C15
0.1uF
GND
POWER SIGNALS
VCC_5V
VBAT
GND
VCC_5V
VBAT
GND
CHARGER OUTPUT
5V_USB
J?
MH1
1
2
3
4
USB_GND1
MH2
901-141A1011D10100
GND
5V_USB
J?
MH1
1
2
3
4
USB_GND2
MH2
901-141A1011D10100
GND
PROGRAMMING USB
J?
U241-041N-1BR85-1
MH1
1
2
RX
3
TX
4
MH2
GND
GND
PROGRAMMING USB
LED_IN
LED?
R?
100mohs
2835 SMD LED
LED?
R?
100mohs
2835 SMD LED
LED?
R?
100mohs
2835 SMD LED
LED?
R?
100mohs
2835 SMD LED
GND
DC POWER HEADER
CHRG_5V
J?
1
3
2
DC-C024E-7A-1.65
GND
ISP PROGRAMMING HEADER
VCC_5V
P1
D12/MISO
D13/SCK
D11/MOSI
RESET
1
2
3
4
5
6
61300611121
GND
BATTERY HEADER
VBAT
P2
R7
0.001R
IN
D2
LESD5Z5.0CT1G
1
2
826629-2
GND
GND
IN signal comes from Current sense amplifier circuitry

DECOUPLING CAPACITOR



5V BOOST REGULATOR



RESISTOR DIVIDER

